LOVELY PROFESSIONAL UNIVERSITY

Sample Question Set - Unit 4

Course Code: INT396

Course Title: UNSUPERVISED LEARNING

Course Outcome: CO4

Paper: Sample Multiple Choice Questions

Maximum Marks: 40 x 1 = 40

Unit: 4

Unit Title: Dimensionality Reduction and Representation Learning

Unit Mapped RBT: L5 (Evaluate)

Total Questions: 40

Suggested Duration: 60 Minutes

Answer Grid (to be filled by the student)

Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
Q1		Q11		Q21		Q31	
Q2		Q12		Q22		Q32	
Q3		Q13		Q23		Q33	
Q4		Q14		Q24		Q34	
Q5		Q15		Q25		Q35	
Q6		Q16		Q26		Q36	
Q7		Q17		Q27		Q37	
Q8		Q18		Q28		Q38	
Q9		Q19		Q29		Q39	
Q10		Q20		Q30		Q40	

Multiple Choice Questions (40 x 1 = 40 Marks)

Q1. Name the matrix whose eigenvectors give the principal components of a dataset.

A) The covariance matrix
B) The proximity matrix
C) The confusion matrix
D) The adjacency matrix

Q2. Identify what an eigenvalue of the covariance matrix reports about its component.

A) The number of records in that direction
B) The variance captured along that direction
C) The angle between two components
D) The reconstruction error of the component

Q3. State the part of an autoencoder that maps the input down to the bottleneck layer.

A) The decoder
B) The classifier
C) The encoder
D) The optimiser

Q4. Name the quantity an autoencoder is trained to minimise.

A) The classification accuracy
B) The explained variance
C) The silhouette score
D) The reconstruction error

Q5. Identify the t-SNE setting that governs how many neighbours each point is balanced against.

A) Learning rate
B) Perplexity
C) Batch size
D) Epsilon

Q6. State the largest number of principal components that a dataset of p features can produce.

A) p - 1
B) 2p
C) Two
D) p

Q7. Explain why the data must be centred before the principal components are computed.

A) Variance is defined as spread about the mean
B) Centring removes the outliers from the table
C) Centring makes every feature take the same units
D) Centring guarantees the components are orthogonal

Q8. Distinguish an undercomplete autoencoder from an overcomplete one.

A) Undercomplete has a bottleneck wider than the input
B) Undercomplete uses no decoder at any stage
C) Undercomplete has a bottleneck narrower than the input
D) Undercomplete is trained on labelled data only

Q9. Describe what the manifold hypothesis claims about high-dimensional data.

A) It fills the whole of the high-dimensional space
B) It always contains exactly two natural clusters
C) It lies near a surface of far fewer dimensions
D) It cannot be reduced without losing everything

Q10. Interpret an explained variance ratio of 0.62 reported for the first principal component.

A) It captures 62 per cent of the total variance
B) It is correlated 0.62 with the second component
C) It reconstructs 62 per cent of the records exactly
D) It requires 62 per cent of the original features

Q11. Predict what happens to a t-SNE layout when the perplexity setting is changed.

A) The layout is completely unaffected by it
B) The records are removed from the plot entirely
C) The axes acquire meaningful units of measure
D) The apparent number and shape of the groups can change

Q12. Recognize why t-SNE cannot place a new record onto an existing plot without being refitted.

A) It discards the original features after fitting
B) It learns a layout, not a reusable mapping
C) It requires the record to carry a class label
D) It can only handle exactly two dimensions

Q13. Explain the sense in which Principal Component Analysis is an unsupervised technique.

A) It requires a label for every record used
B) It predicts a numeric target for each record
C) It measures accuracy against known answers
D) It uses the features alone and never a target

Q14. Describe what a denoising autoencoder is trained to do.

A) Rebuild a record from its class label alone
B) Remove every outlier from the training table
C) Rebuild a clean record from a corrupted copy
D) Compress the data without any decoder at all

Q15. Explain why a linear autoencoder trained with a squared-error loss recovers the same subspace as PCA.

A) Both are trained on the class labels of the data
B) Both find the best linear subspace for reconstruction
C) Both require the features to be uncorrelated first
D) Both retain every one of the original features

Q16. Predict the effect of a single extreme outlier on the first principal component.

A) It can pull the component towards itself
B) It is automatically excluded by the method
C) It has no effect because of the centring step
D) It causes the component to become undefined

Q17. Solve the following: five components explain 40, 25, 15, 12 and 8 per cent of the variance. Determine the smallest number retaining at least 80 per cent.

A) 2
B) 3
C) 4
D) 5

Q18. Compute the cumulative variance explained by the first two of four components explaining 55, 20, 15 and 10 per cent.

A) 75 per cent
B) 55 per cent
C) 20 per cent
D) 90 per cent

Q19. Apply the Kaiser criterion to the eigenvalues 3.2, 1.8, 1.1, 0.7 and 0.2, and determine how many components are retained.

A) 2
B) 4
C) 5
D) 3

Q20. Solve the following: a dataset of 500 features is reduced to 25 components. Determine the reduced size as a fraction of the original.

A) One fifth
B) One tenth
C) One twentieth
D) One fiftieth

Q21. Use the definition of reconstruction error to determine what a value of exactly zero would mean.

A) The decoder reproduces every input exactly
B) The bottleneck holds no information at all
C) The network has not yet been trained once

D) The inputs were all identical to each other

Q22. Compute the variance explained by the fourth component when the first three together explain 61 per cent and the first four explain 69 per cent.

A) 4 per cent
B) 61 per cent
C) 8 per cent
D) 69 per cent

Q23. Apply Principal Component Analysis to a dataset of 12 features and determine how many components are produced when no limit is set.

A) 11
B) 12
C) 6
D) 2

Q24. Examine the following and determine the consequence: Principal Component Analysis is applied to raw columns holding height in metres alongside salary in rupees.

A) The height column dominates the first component
B) The two columns contribute equally to the result
C) The components come out perfectly uncorrelated
D) The salary column dominates the first component

Q25. Use the following explained variances to determine how many components reach a 95 per cent threshold: 70, 18, 7, 3 and 2 per cent.

A) 2
B) 4
C) 3
D) 5

Q26. Compare Principal Component Analysis with t-SNE on the ability to transform a record that was not in the training set.

A) t-SNE can transform it, PCA cannot
B) Both of the two can transform it directly
C) Neither of the two can transform it at all
D) PCA can transform it, t-SNE cannot

Q27. Evaluate the claim that a principal component explaining very little variance can always be discarded safely.

A) It is unsafe, since low variance is not low usefulness
B) It is always safe, as variance measures usefulness
C) It is always safe, since components are uncorrelated
D) It is unsafe only when the data is standardised

Q28. Assess the reading of a t-SNE plot in which the gap between two islands is taken as a measure of how different the two groups are.

A) The reading is sound, as t-SNE preserves all distances
B) The reading is unsound, as global distances are not preserved
C) The reading is sound provided the data was scaled
D) The reading is unsound only for very small datasets

Q29. Justify the use of Principal Component Analysis before clustering a table of 800 heavily correlated features.

A) It supplies the number of clusters to be used
B) It guarantees that the clusters will be spherical
C) It replaces the need to validate the clustering
D) It removes redundancy and restores contrast in the distances

Q30. Determine the more suitable approach when every reduced feature must be traceable to a quantity that was actually measured.

A) Principal Component Analysis
B) Feature selection
C) An autoencoder bottleneck
D) A t-SNE embedding

Q31. Recommend the correct response to a scree plot that falls smoothly with no visible elbow.

A) Retain exactly two components in every case
B) Abandon dimensionality reduction altogether
C) Choose by a variance threshold and report the choice
D) Retain every component the method produced

Q32. Judge whether a very low reconstruction error proves that an autoencoder representation will serve the task that follows.

A) It does not, since faithful is not the same as useful
B) It does, because the information is provably intact
C) It does, provided the bottleneck is narrow enough
D) It does not, because the error is always unreliable

Q33. Measure the share of the original information a Principal Component Analysis projection retains by selecting the appropriate quantity.

A) The cumulative explained variance ratio
B) The silhouette coefficient of the projection
C) The number of components produced
D) The Davies-Bouldin Index of the result

Q34. Assess the risk of applying Principal Component Analysis when the informative signal has a small variance.

A) The signal is always placed in the first component
B) The method refuses to run in that situation
C) The signal is amplified by the projection step
D) The signal may be dropped with the discarded components

Q35. Evaluate the claim that UMAP should always be preferred to t-SNE.

A) UMAP is always better because it is faster
B) Neither is always better; the choice depends on the task
C) t-SNE is always better because it is older
D) The two produce identical layouts in every case

Q36. Recommend what must accompany a two-dimensional embedding placed in a report.

A) The number of clusters the analyst expected
B) The accuracy the embedding achieved on test data
C) The method, its settings and the random seed used
D) Nothing further, since the picture speaks for itself

Q37. Determine the representation to prefer when the finished model must run on a device with very little memory.

A) Every one of the original raw features
B) A two-dimensional t-SNE layout of the data
C) A small number of components or a narrow bottleneck
D) The full proximity matrix of the records

Q38. Critique a project that reports 99 per cent of the variance retained while the clustering that followed became noticeably worse.

A) The variance figure must have been miscalculated
B) Retained variance is not evidence of retained structure
C) Clustering always degrades after any reduction
D) The project should have retained 100 per cent

Q39. Judge the claim that dimensionality reduction improves the performance of any method it is applied to.

A) True, because noise is always removed by it
B) True, because fewer features always help
C) False, because it never changes performance
D) False, since useful information can be discarded too

Q40. Evaluate the decision to train an autoencoder rather than run Principal Component Analysis on a dataset holding only 300 records.

A) PCA is the better choice, as the network would overfit
B) The autoencoder is better, as it is always stronger
C) The two would give identical results at this size
D) Neither method can be applied to 300 records

----- End of Sample Question Set -----